

KabadiAI: Project Plan

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1. The idea

KabadiAI is a multilingual mobile web app. Anyone with a drawer of dead gadgets photographs a pile and learns three things in seconds: what the items are and which are dangerous, roughly what they are worth, and the safest place to take them. The answer arrives in the person's own language, read aloud where a voice exists, and ends with a shareable card they can show or send to any scrap dealer to check that an offer is fair. An optional handoff slip records that a pile went to an authorized recycler.

The one-line pitch: know what your e-waste is, what it is worth, and where it should go.

2. The problem

- India generates over 3 million tonnes of e-waste a year, and over 90% is processed by the informal sector (peer-reviewed case study on kabadiwalas, ScienceDirect, 2023).
- About 45% of surveyed people prefer to sell e-waste to kabadiwalas (same study).
- What informal handling does not recover is often landfilled or burned, which releases toxic fumes.
- Under the E-Waste (Management) Rules, 2022, producers meet targets by buying EPR certificates, and only registered recyclers can generate them. One policy analysis notes the informal sector draws no recognition in the rules, while the government states a goal to channel the informal sector into the formal one.
- The gap: households cannot tell what is hazardous, what is worth money, or where an authorized recycler is. Collectors work mostly in cash, and one older study found only about 30% owned a smartphone.

So the app targets the household, who has the phone and the choice. The dealer is reached through a share card, so no smartphone or digital payment is needed on that side. We do not pay anyone, and we do not claim any subsidy exists.

3. Users

- Meera, a flat-dweller in Delhi-NCR with old chargers, a swollen phone battery and a dead laptop. Wants to know what to do and not get cheated.

- Ramesh, a local scrap dealer. Does not need the app. Receives a WhatsApp card showing the item list and a fair range.
- A recycler. Can open a handoff slip to see what is arriving.

4. User journey

1. Choose a language (Hindi, English, Punjabi, Bengali, Tamil, Telugu, Marathi at launch).
2. Tap the big camera button and photograph the pile.
3. The app shows a progress screen, then a result: item list with condition, a red banner for any hazard, an indicative value range, and a confidence level.
4. If confidence is low, the agent asks for a closer or clearer photo instead of guessing.
5. The hazard warning is read aloud. Example: do not burn, puncture or crush a swollen lithium battery.
6. Fair-price check: the user types an offer from a dealer, and the app says low, fair or high against the indicative range.
7. Where to take it: nearest authorized recyclers with distance, address and a directions link.
8. Share: a card image with items, range and hazards, sent over WhatsApp.
9. Optional handoff slip with an ID and QR. Scanning shows items and status.

5. Ideas borrowed from other projects

Source	What we take
First Commit winners (Suraksha, Beacon Night Shift, CampusEvac)	An agent that acts for one specific person, voice as a first-class feature, a safety-first UI, and a live deployed link
Fix-Finder (1st place, Henkel Hackathon PH 2026)	Photo in, practical advice out
Husker AI (Pan-African Hack-AI-Thon 2026)	Turn waste into value by connecting it to a buyer
EcoBin (Global Appathon 2026)	Visible impact tracking
KisanSaathi and Oikos (Global Appathon 2026)	Simple, mostly offline interfaces
TRACE (AI x City Climate 2026)	A credible pilot path in the writeup

Source	What we take
GitLab AI Hackathon, DocSync	Act only when confident, otherwise ask for human input. Becomes our confidence gate
GitLab, Green Agent category	Measure footprint honestly, with labelled estimates
Judges at JetBrains and Devpost	One thing done really well, and the demo video is the whole pitch

Existing apps such as Scrapdeal, ScrapUncle, Recykal and Karo Sambhav focus on pickup logistics. We focus on identification, hazard safety, fair-price transparency and local-language guidance. Verify this positioning before the demo and never claim others lack a feature unless confirmed.

6. Scope

P0 (must work): photo to items, hazards, indicative value, recycler list, fair-price check, share card, six-plus languages for the interface.

P1 (only after P0 is solid): handoff slip with QR, confidence follow-up flow polish, impact counter with clearly labelled estimates, offline queue.

P2 (cut first): speech input, history sync across devices, extra cities.

Will not build: payments, UPI, EPR subsidies, tokens, accounts and logins, kiosk hardware, native apps.

7. Frontend plan

Stack: Next.js with Tailwind, built as a PWA (manifest plus service worker), hosted on AWS Amplify Hosting. Mobile-first with a max width so it looks like a phone screen on desktop.

Screens: language picker, home, scanning, result, clarify (retake), fair-price check, where to take it, share card, handoff slip (P1), local history.

Design rules:

- Big camera button, large touch targets, minimal text.
- Green means safe or money, red means hazard, amber means unsure. Icons carry meaning, text backs them up.
- Every screen works with sound off, and every result has a replay-voice button.
- Test at 360 px width on a mid-range Android.

Technical details:

- Camera: an input tag with accept image and capture environment opens the native camera.
- Compress the photo in the browser to about 1280 px before upload. Lambda payload limits are around 6 MB, and base64 adds roughly a third. Use a presigned S3 upload for larger images (verify limits).
- Location: browser geolocation, with a city or pincode picker as fallback.
- Share card: draw to a canvas, export as an image, use the Web Share API where available, otherwise a wa.me link.
- QR: a small client library, P1 only.
- Offline: queue the photo in IndexedDB and show a clear message. Process when back online.
- Voice: Polly audio for Hindi and Indian English. For other languages use the browser speech engine if a voice is installed, else text and icons only.
- Strings: one JSON file per language, loaded on demand.

8. Backend plan

Deployment: AWS SAM template (an AWS open-source tool) defining everything, so the stack is reproducible and visible in the video.

Endpoints (Lambda behind API Gateway or a Function URL):

- POST /scan: photo reference in, structured result out.
- GET /recyclers: nearest authorized recyclers for a latitude and longitude.
- POST /speak: text and language in, Polly audio URL out (cached).
- POST /translate: dynamic text through Amazon Translate (cached).
- POST /handoff and GET /handoff/{id}: P1.

The agent (Strands Agents SDK):

- Model: Amazon Nova on Bedrock now. Switch to Claude on Bedrock if access is approved. Provider is an environment variable.
- Tools: analyze_image, request_clarification, fetch_indicative_price, find_recyclers, build_share_summary.
- Flow: analyze, then if confidence is below 70 ask for a better photo, otherwise price each item, find recyclers, compose the result.
- Facts come from tables, not the model: prices from DynamoDB, hazard messages from a fixed reviewed table, recyclers from the seeded list.
- Output is validated against a schema, with one retry on bad JSON.

Data (DynamoDB):

- Prices: component, min and max rupees per kg, source note, date.
- Recyclers: id, name, address, city, coordinates, authorization source and date checked.
- Scans: id, timestamp, result, TTL (auto-delete). No names or phone numbers.
- Handoffs (P1): id, items, status, timestamps.
- TranslationCache: key, language, text.

Other services:

- S3 for images, with a lifecycle rule that deletes them within a day.
- SNS to notify a recycler contact by email in the demo.
- Polly for voice, Translate for text.
- CloudWatch dashboard showing requests, latency and errors, which also serves as AWS evidence in the video.

Protection: billing alert, API throttling, a simple request token, and least-privilege IAM roles. Keys never go in the repo.

9. Languages

The model returns language-neutral codes. Everything shown to the user is looked up by language.

- Amazon Translate supports Hindi, Bengali, Gujarati, Kannada, Malayalam, Marathi, Punjabi, Tamil, Telugu and Urdu.
- Polly neural voices cover Hindi and Indian English among Indian options. Other languages use device voices or text.
- Safety messages are short fixed strings, translated once and checked by a native speaker. They never go through live machine translation.
- Component names come from a lookup table.
- Only free-text advice goes through Translate.
- Adding a language means adding one JSON file.

10. Safety, privacy and honesty

- Prices are always labelled indicative, with a date and source note.
- Hazard guidance is conservative and fixed. The app is not legal or medical advice.
- Recycler data covers Delhi-NCR only, from the DPCC and CPCB approved list (dated 2023), re-verified before the demo. Say so on screen.
- Images are deleted quickly, scans store no personal data, and the app asks for location only when needed.

- Impact numbers, if shown, are labelled estimates.

11. Testing and proof

1. Collect 40 or more real photos of e-waste in varied light, mixed piles and single items.
2. Write ground truth by hand for each photo: components, condition, hazards.
3. Measure component accuracy, hazard recall (the most important number), confidence calibration and latency.
4. Compare Nova Lite and Nova Pro, and Claude if access arrives. Keep the best.
5. Put the table in the README and show it in the video. Report honestly, including failures.
6. Get one real scrap dealer to confirm price ranges, and ask what phone he uses.

12. Schedule and roles

Suggested split, adjust to taste: Vivek takes backend, agent and AWS. Souhardya takes frontend, languages and the share card.

- **Thu Oct 8:** new public repo, SAM skeleton, scan pipeline photo to JSON on Lambda, first deploy by evening.
- **Fri Oct 9:** hazards table, prices, recyclers, fair-price check, frontend result screens.
- **Sat Oct 10:** languages and voice, share card, accuracy test, UI polish. Optional DTU build day for mentor feedback.
- **Sun Oct 11:** feature freeze by noon. Record video, finish README and writeup, submit. Submissions open at 8 AM, so confirm the deadline on the schedule page.

Commit small and often with honest messages. The repo history must match the event dates.

13. Demo video and submission

Video, under 3 minutes, on YouTube (public or unlisted, tested signed out):

- 0:00–0:25 the dealer quote and the problem.
- 0:25–1:45 live phone demo: photo, hazard voice, value range, fair-price check, recycler list, WhatsApp card. Show one language switch.
- 1:45–2:30 AWS: SAM stack, Lambda, Bedrock, DynamoDB, CloudWatch on screen.
- 2:30–3:00 accuracy table, limits, and next steps.

Submission: public repo, video, writeup (problem, build, where AWS fits, AI tools used, limitations). Also publish a short blog on AWS Builder Center.

14. Rules checklist

- Student status verified on Builder Center for every member.
- Every member registered with their own account.
- Nothing built before the clock starts. Practice material stays out of the repo.
- Uses AWS and shows it in the video.
- AI tools (Claude, Antigravity) listed in the writeup.
- Third-party material credited with licences.
- Video under 3 minutes and public or unlisted.

15. Risks and fallbacks

- Model misreads parts: try the stronger model, tighten the prompt, narrow the component list.
- Claude access on Bedrock stays blocked: Nova already covers us.
- Voice sounds wrong in a language: show text and icons.
- Deployment trouble: use a container image for Lambda, or ask mentors early.
- Out of time: cut in the order P2, impact counter, handoff slip. Never cut the core flow or the video.

16. After the hackathon

Pilot with one authorized recycler and a few scrap dealers in Delhi-NCR, add more cities from CPCB lists, and explore a way for dealers to confirm fair-price cards. These are next steps only and are not claimed as built.